EDMS 03-300-2

SPECIFICATION

FOR

UNDERGROUND XLPE CABLES 0.6/1(1.2)KV

ALUMINUM / COPPER, ARMOURED

Issue: Jul-2020/ Rev- 2

EDMS 03-300-2

SPECIFICATION

FOR

UNDERGROUND XLPE CABLES 0.6/1(1.2)KV

ALUMINUM / COPPER, ARMOURED

Issue: Jul-2020/ Rev- 2

توجد بنود اختيارية (Option items) يجب تحديدها بواسطة شركة التوزيع قبل الطرح.

CONTENTS

1. SCOPE

This specification covers the design, construction, testing and supply of low voltage cables U_0/U = 0.6/1 kV, 50 Hz multi cores with stranded compact AL/CU conductors , extruded XLPE insulation, double tape armour and PVC over sheath. The cables should be used in system with solidly earthed neutral. Th cables should be for underground direct buried with cross section areas as follow:

3x500 + 240 mm^2
3x400 + 240 mm^2
3x300 + 150 mm^2
3x240 + 120 mm^2
3x185 + 95 mm^2
3x150 + 70 mm^2
3x120 + 70 mm^2
3x95 + 50 mm^2
3x70 + 35 mm^2
3x50 + 25 mm^2
3x35 + 16 mm^2
4x25 mm^2

2. APPLICABLE STANDARDS:

Unless otherwise specified in this specification, the L.V cables should comply with the latest edition of IEC standard and ISO codes of standards as following:

S. N.o	Standard No.	Description
1	IEC 60502-1	Power cables with extruded insultation and their acc. For rated voltages from 1 KV (U_m =1.2 KVand 3 KV (U_m = 3.6 KV)
2	IEC 60811	Test methods for non-metallic materials
3	IEC 60228	Conductors of insulated cables
4	IEC 60060	High-voltage test techniques
5	IEC 60287	Calculation of the current rating

3. ENVIRONMENTAL CONDITIONS

The performance of the extruded XLPE insulated cables should be guaranteed for following environmental conditions, any differences in the guaranteed performance should be clearly set out in the offer.

Minimum ambient temperature : -5°C.
Maximum ambient temperature : 45°C (50 °C as option).
Maximum relative humidity : 95%.
Maximum altitude : 1000 m.

4. SYSTEM DETAILS :

Frequency	50 Hz
Number of phases	Three
Nominal voltage	0.4 kV
Highest voltage	0.6 kV
Maximum S.C. temperature	250°C

5. REQUIREMENTS FOR CONSTRUCTION

5.1 CONDUCTORS

The cables should be have non-compacted circular class 2 plain AL or CU stranded wires. The conductors should be circular or sector-shaped. The minimum number of wire strands in the conductor should be as following table.

No.	Conductor C.S.A (mm^2)	No. of wires	Max. DC resistance at 20°C (Ω/Km) AL	Max. DC resistance at 20°C (Ω/Km) CU
1	16	7	1.91	1.15
2	25	7	1.2	0.727
3	35	7	0.868	0.524
4	50	19	0.641	0.387
5	70	19	0.443	0.268
6	95	19	0.320	0.193
7	120	37	0.253	0.153
8	150	37	0.206	0.124
9	185	37	0.164	0.0991
10	240	61	0.125	0.0754
11	300	61	0.1	0.0601
12	400	61	0.0778	0.047
13	500	61	0.0605	0.0366

5.2 INSULATION

5.2.1 Nominal Insulation thickness should be not less than the specified values given in the following table.

No.	Conductor C.S.A (mm^2)	Nominal insulation thickness (mm)
1	16	0.7
2	25	0.9
3	35	0.9
4	50	1
5	70	1.1
6	95	1.1
7	120	1.2
8	150	1.4
9	185	1.6
10	240	1.7
11	300	1.8
12	400	2
13	500	2.2

Note: the lowest value of thickness of insulation at any point should be not fall below the nominal value by more than 10%of this value +0.1mm.

5.3 ASSEMBLY OF CORES, INNER COVERING AND FILLERS

5.3.1 Multi core cables with armour should have an inner covering over the laid up cores.

5.3.2 The inner covering may be extruded.

5.3.3 The inner covering and fillers should be made of suitable materials. An open helix of Suitable tape is permitted as a binder before application of an extruded inner covering.

5.3.4 The material used for inner coverings and fillers should be suitable for the operating temperature of the cable and compatible with the insulating material. The thickness of extruded inner covering (nominal value) should be derived from the following table.

Fictitious diam. Over laid-up cores		**Thickness of extruded inner coverings (nominal values)**
Above	**Up to and including**	
mm	**mm**	**mm**
--	**25**	**1.0**
25	**35**	**1.2**
35	**45**	**1.4**
45	**60**	**1.6**
60	**80**	**1.8**
80	**--**	**2.0**

5.3.5 The thickness of extruded inner covering (nominal values) should eb as follow:

Cable C.S.A (mm^2)	thickness of inner covering (mm)
3x500+240 mm^2	1.8
3x400+240 mm^2	1.8
3x300 + 150 mm^2	1.6
3x240 + 120 mm^2	1.6
3x185 + 95 mm^2	1.4
3x150 + 70 mm^2	1.4
3x120 + 70 mm^2	1.4
3x95 + 50 mm^2	1.2
3x70 + 35 mm^2	1.2
3x50 + 25 mm^2	1
3x35 + 16 mm^2	1
4x25 mm^2	1

Note: Inner covering thickness: mean value should be $\geq$ value corresponding to cable size in this table of above specifications, the lowest value of thickness at any point should not fall below table value by more than 20 % of this value + 0.2 mm.

5.4 ARMOUR

5.4.1 The armour should be applied on inner covering complying with clause 5 of IEC 60502-1.

5.4.2 The armour should be of cold rolled steel tape. The thickness of tape should be derived from the following table.

Fictitious diam. under the armour		Thickness of steel tape
Above	Up to and including	
mm	mm	mm
--	30	0.2
30	70	0.5
70	--	0.8

5.4.3 The armour tape should be applied helical in two layers so that the outer tape is approximately central over the gap of the inner tape which should not exceed 50% of the tape width.

5.4.4 Thickness of armour tape should eb as following table:

Cable C.S.A (mm^2)	Thickness of armour steel tape (mm)
3x500+240 mm^2	0.5
3x400+240 mm^2	0.5
3x300 + 150 mm^2	0.5
3x240 + 120 mm^2	0.5
3x185 + 95 mm^2	0.5
3x150 + 70 mm^2	0.5
3x120 + 70 mm^2	0.5
3x95 + 50 mm^2	0.5
3x70 + 35 mm^2	0.2
3x50 + 25 mm^2	0.2
3x35 + 16 mm^2	0.2
4x25 mm^2	0.2

5.5 NON-METALLIC SHEATH

5.5.1 The sheath should be P.V.C. compound suitable for the cable operating temperature.

5.5.2 The sheath should be applied directly over the armour.

5.5.3 The nominal thickness of sheath should be not less than values here in the following table.

Cable C.S.A (mm^2)	Nominal (average) thickness of sheath (mm)
3x500+240 mm^2	3.5
3x400+240 mm^2	3.3
3x300 + 150 mm^2	3
3x240 + 120 mm^2	2.8
3x185 + 95 mm^2	2.6
3x150 + 70 mm^2	2.5
3x120 + 70 mm^2	2.3
3x95 + 50 mm^2	2.2
3x70 + 35 mm^2	2
3x50 + 25 mm^2	1.9
3x35 + 16 mm^2	1.8
4x25 mm^2	1.8

6. GENERAL REQUIREMENTS:

6.1 CORE IDENTIFICATION

- The cores should be identified by colors. Each core of cable should be achieved by using colored insulation. The three cores should be red, yellow and light Blue.
- The neutral core should be black.

6.2 IDENTIFICATION OF CABLE PHASES AND MARKING

The cable to which this specification relates to should be identified on the outer sheath with the following items (data):

- Rated voltage.
- Size (number of cores*C.S.A of conductor mm^2).
- Material of conductor.
- Insulation materials.
- Manufacturer’s name.
- Year of manufacture.

- Customer name (…EDC)
- Cable length.

Distance between marking should be not exceed one meter and each mark should serially indicate the remaining length of cable.

6.3 MISCELLANEOUS

1- The cable length on each reel should be as follows:

Cable C.S.A (mm^2)	Cable length on reel (m)±5%
3x500+240 mm^2	500AL/300CU,
3x400+240 mm^2	500 AL & 350 CU
3x300 + 150 mm^2	500 AL/CU
3x240 + 120 mm^2	500 AL/CU
3x185 + 95 mm^2	600 AL/CU
3x150 + 70 mm^2	600 AL/CU
3x120 + 70 mm^2	800 AL/CU
3x95 + 50 mm^2	800 AL/CU
3x70 + 35 mm^2	1000 AL/CU
3x50 + 25 mm^2	1000 AL/CU
3x35 + 16 mm^2	1000 AL/CU
4x25 mm^2	1000 AL/CU

2- All cable ends should be firmly secured to the reel and should be covered by heat-shrinkable end caps.

3- The cable reels should be covered with wooden slabs which tighted with steel tap of suitable thickness.

4- All cables should be supplied on reels whose size not to exceed 2.6 m diameter and 1.6 m width including protrusion bolts length. The reels may be returnable as per agreement between the tender and …EDC.

5- ….EDC has the right to change the cable length and consequently the reel diameter will be changed to max size 3 m.

6- On each reel the following data should be printed on both sides and could not be erased easily:

- For ….EDC: ARAB REPUBLIC OF EGYPT
- LOW VOLTAGE POWER CABLES:
- MANUFACTURED BY :
- VOLTAGE CLASS : KV
- NUMBER OF CORES & CROSS-SECTION AREA : mm^2
- MATERIAL OF CONDUCTORS :
- TYPE OF INSULATION :
- LENGTH OF CABLE : M
- GROSS WEIGHT : KG
- NET WEIGHT : KG
- REEL NO/ DAT OF MANUFACTURE

7. TESTS

7.1 GENERAL

- All cables shall be tested in accordance with the latest standards and as specified herein. Supplier shall provide all test results for review and acceptance by EEHC
- The full range of Routine, Sample and Type tests specified in IEC 60502-1, clause 14, 15, 16 and 17 shall be carried out as applicable.
- Routine and/or special tests should be carried out in the supplier's factory.

7.2 TESTS CONDITIONS

- Voltage tests should be made at ambient temperature 20 ± 15ºC and other tests at ambient 20 ± 5ºC
- The frequency of alternating test voltage should be in the range 49–50 HZ.

7.3 ROUTINE TESTS:

1. ELECTRICAL RESISTANCE OF CONDUCTORS.

a- The measurement should be made on all the conductors of each cable length.

b- The complete cable length or sample should be in the test room maintained at reasonably constant temperature for at least 12 h before the test. Alternatively the resistance can be measured on a sample of conductor conditioned for at least 1h in a temperature – controlled oil bath. The measured value of resistance should be corrected to a temperature of 20ºC and 1 km length in accordance with the formula and factors in clause 5 of IEC 60228.

2. HIGH VOLTAGE TEST

a- The voltage test should be made at ambient temperature, using either alternating voltage at power frequency or direct voltage.

b- The test voltage should be applied for 5 minutes in succession between each insulated conductor and all other conductors and metallic cover if any.

c- Test voltage

The test voltage r.m.s should be 3.5 kv. If for three core cables, the voltage test is carried with three phases transformer, the test voltage between the phases should be 1.72 times the given value above.

d- When direct voltage is used, the applied voltage should be 2.4 times the power frequency test voltage.

Requirement: No break down of insulation should occur.

7.4 <u>TAYPE TESTS:</u>

a- Insulation resistance measurement at ambient temperature.

b- Insulation resistance measurement at maximum conductor temperature in normal operation.

c- Voltage test.

The cores of the cable sample shall be immersed in water at ambient temperature for at least 1 hour before the test. A power frequency voltage equal to 4 U0 shall then be gradually applied and maintained continuously for 4 hours between each conductor and the water

7.5 <u>SPECIAL TESTS:</u>

7.5.1 <u>GENERAL.</u>

The special tests required are:

a- Conductor examination.

b- Check of dimensions.

7.5.2 FREQUENCY OF SPECIAL TESTS

One sample per each ten km's of cables

7.5.3 CONDUCTOR EXAMINATION

Compliance with the requirements of IEC 60228 should be checked by inspection and by measurement when practicable.

7.5.4 MEASUREMENT OF THICKNESS OF INSULATION AND OF NON-METALLIC SHEATH.

7.5.4.1GENERAL

Each cable length selected for the test should be represented by a piece of cable taken from one end after having discarded, if necessary. If the average thickness measured or the lowest value measured fails to meet the requirements specified below two other pieces should be checked.

If both of further pieces meet the specified requirements the cable is deemed to comply, but if either does not meet the requirements, the cable is deemed not to comply.

7.5.4.2 TEST METHOD

The test method should be in accordance with IEC 811 as follows:

a- Measuring Equipment

A measuring microscope allowing a reading of 0.01 mm. Instead of the microscope, a suitable measuring projector with magnification power of at least 10 may be used. In case of doughty microscope should be used.

b- Preparation of test pieces any covering should be removed from the insulation and the conductor should be withdrawn, care being taken to avoid damage of insulation.

Each test piece should consist of a thin slice of insulation. The slice should be cut with a suitable device (sharp knife or razor blade) along a plane perpendicular to the axis of conductor.

If a marking is stamped into the insulation or sheath thus giving rise to the local reduction of thickness the test piece should be taken so as to include such marking.

c- Measuring procedure

The test piece should be placed under the measuring equipment with the plane of cut perpendicular to the optical axis.

1- When the inner profile of the test piece is a circle, six measurement should be carried out radially equally spaced around the circumference.

2- For sector-shaped cores, the six measurements should be made as shown in fig.1 attached.

3- When the insulation is taken from a stranded conductor, six measurements are to be made radially in the positions where the insulation is thinnest i.e. between the ridges caused by strands fig2.

4- When the outer profile shows unevenness, the crosswise of the microscope should be adjusted as shown in fig 3.

5- If the inner substantially circular surface is not regular or smooth, the six measurements should be made radially in the positions where the sheath is thinnest placing the cross-wire of the microscope as shown in fig 4.

6- When the inner profile is not circular, an appropriate number of measurements (up to six) should be carried out radially where the sheath is thinnest i.e. at the bottom of the grooves caused by the cores as shown in fig 5.

7- The first measurement should be made at the place where the insulation is thinnest.

7.5.4.3 REQUIREMENTS

a- Insulation

For each piece of core the average of the measured values, rounded to 0.1 mm, should be not less than the specified nominal thickness, and the smallest value should not fall below the nominal value by more than 0.1 mm + 10% of the specified nominal value i.e $\mathbf{t}_m > \mathbf{t}_n - (0.1+0.1\mathbf{t}_n)$ in millimeters where:-

$\mathbf{t}_m$ = minimum thickness

$\mathbf{t}_n$ = nominal thickness

b- Non-metallic sheaths

The piece of sheath should comply with the following.

- For a sheath applied on a smooth cylindrical surface (ex. On an inner covering, a metal sheath, or the insulation of a single core). The average of the measured values, rounded to 0.1 mm should be not less than the specified nominal thickness and the smallest value measured should not fall below the specified nominal value by more than 0.1 mm + 15 % of the specified nominal value.

- For a sheath applied on an irregular cylindrical surface (ex. a penetrating sheath on an unarmoured multicore cable without inner covering or a sheath applied directly over armour, metallic screen, or concentric conductor) and for a separation sheath, the smallest measured value should not fall below the specified nominal thickness by more than: 0.2 mm + 20 % of the specified nominal thickness.

<u>IMPORTANT NOTE</u>

- The factory should be approved by the Egypt Electricity Holding Company.
- Type test report/certificate from EEHC approved independent testing laboratory should be submitted to …EDC.

- The following guarantee table should eb filled by the tenderer and submitted to …EDC

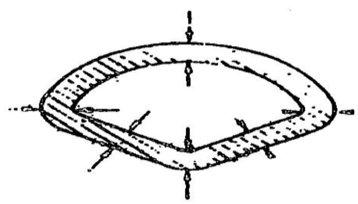

FIGURE 1

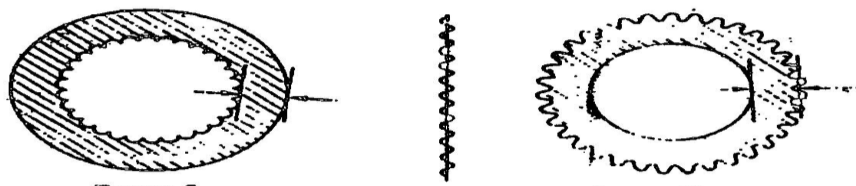

FIGURE 2

FIGURE 3

FIGURE 4

FIGURE 5

Different Figures of Measuring Procedure

GUARANTEE TABLE FOR CABLE ….X…..+….mm^2

A - TECHNICAL PARTICULARS:-	
- Rated voltage	…………………………kv
- Manufacturer's name	…………………………
- Type of cable	…………………………
- Standards specifications applied to cable	…………………………
B – conductor data:-	
- Number of cores / cable	…………………………
- Nominal Cross-sectional area /core	…………………………mm^2
- Number of strands per core / diameter of strand (before compaction)	…………………………mm
- Cross sectional area of each strand	…………………………mm^2
- Guaranteed cross sectional area / cond	…………………………mm^2
- Over all diameter of conductor	…………………………mm
- Weight of conductor / km	…………………………kg
- DC Resistance of conductor at 20ºC / km	…………………………ohm
- Temp. Coeff. Of resistance	…………………………
- Current carrying capacity	…………………………amp
C- Insulation & Insulation strength:-	
- Insulation class	…………………………kv
- Material of insulation	…………………………mm
- Thickness of insulation	…………………………
- Min insulation resistance at 95 ºC	…………………………MΩ.km
- Max. allowable final temp. according to cable specification	…………………………ºC
- Test voltage between core and earth:	
a) AC & Time	…………………kv & min
b) DC & Time	……………………kv & min
D – Inner covering:-	…………………………
- material of inner covering	…………………………
- Thickness of inner covering	…………………………mm
E – Armour:-	…………………………

- No of layers	…………………………..
- Thickness of armour	…………………………..mm
F - Outer jacket	…………………………..
- material of outer jacket	…………………………..
- Thickness of outer jacket	…………………………..mm
- External diameter of cable	…………………………..mm
- Weight of cable / drum	…………………………..kg
- Length of cable / drum	…………………………..m
- Diameter of drum	…………………………..m
- Width of drum	…………………………..m
- Gross weight of charged drum	…………………………..kg
- Min. bending radius of cable	…………………………..mm

we guarantee above mentioned data for our afford material for ……EDC ....

vendor / Agent

Date: signature